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GRAPH-ASSISTED DESIGN (GAD): A MULTI-OBJECTIVE GNN–EVOLUTIONARY FRAMEWORK FOR AUTOMATED STRUCTURAL OPTIMIZATION

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Summary. *This work introduces Graph-Assisted Design (GAD), a unified surrogate-assisted framework that encodes structural candidates as attributed graphs and couples a Graph Attention Network (GAT) ensemble with NSGA-III over ten simultaneous objectives. A key contribution is the Surrogate Refresh strategy: periodic high-fidelity 3D FEM evaluations on Pareto-front candidates feed exponential-moving-average correction factors back to bound prediction drift. The framework is applied to compliant multi-material airless wheels (Spring Tires) for planetary rovers, parametrized via an 18-variable genome covering geometry, pantographic spoke configuration, and eight-sector material distribution across FFF-printable polymers. Over 500 generations (population 1 000), a 103-solution Pareto front is obtained. The peak-stiffness candidate (ID 79) reaches $k_v = 796.6$ kN/m and a von Mises safety factor of 20.06 on a 1 328 466-DOF TetP2 solution (scikit-fem v12.0.1), with a Goodman ratio of 0.070 and a shape-recovery ratio of 96.9 %. Limitations are stated explicitly: the mesh-convergence triplet lies outside the asymptotic range, and the stiffness model saturates against an empirical cap, carrying ±30–40 % uncertainty.*

1 INTRODUCTION

Mars surface exploration imposes extreme requirements on rover wheels: temperatures from $-125\,^{\circ}$C to $+20\,^{\circ}$C, abrasive terrain, and no in-situ repair. Conventional pneumatic tyres are impractical, while the aluminium wheels of *Curiosity* suffered crack propagation within years [6]. The Perseverance rover ($m = 1{,}025$ kg, $g_{\mathrm{Mars}} = 3.71$ m/s^2) imposes a dynamic design load of 1 584 N per wheel (static 634 N, dynamic factor 2.5).

The NASA Spring Tire addresses this through a compliant airless wheel: a tread band connected to a rim by interlaced FFF/FDM polymeric spokes acting as 3D springs, enabling tunable stiffness and damping. With six continuous geometric parameters, eight independently selectable material sectors (six polymer grades), and four pantographic spoke variables, the

design space exceeds 10^{15} configurations — beyond classical parametric studies when ten objectives compete simultaneously.

We introduce Graph-Assisted Design (GAD): an automated pipeline combining a GAT-ensemble surrogate, NSGA-III with Surrogate Refresh, and high-fidelity 3D FEM validation (scikit-fem, TetP2).

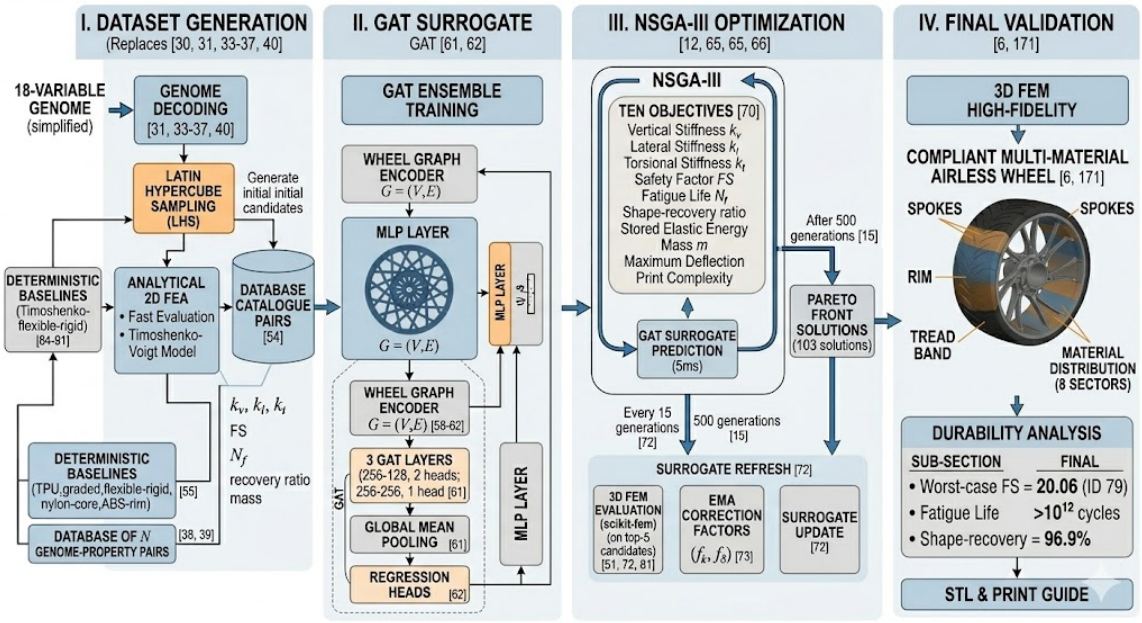


Figure 1: Overview of the GAD pipeline. (I) Dataset generation via LHS and analytical 2D FEA. (II) GAT surrogate ensemble trained on wheel graphs $G = (V, E)$. (III) NSGA-III optimizer with 10 simultaneous objectives and Surrogate Refresh. (IV) High-fidelity 3D FEM validation and STL export.

2 THE 18-VARIABLE DESIGN GENOME

Every candidate is encoded as $\mathbf{g} \in [0,1]^{18}$, decoded by

$$x_i = x_{\min,i} + g_i\,(x_{\max,i} - x_{\min,i}), \quad i = 1, \ldots, 18. \tag{1}$$

Discrete parameters are rounded; material sectors use $\mathrm{mat_id} = \mathrm{clip}(\mathrm{round}(g_i \times 5), 0, 5)$.

Group 1 — Geometry (genes 0–5): outer diameter $d_{\text{ext}} \in [300, 500]$ mm; rim diameter $d_{\text{rim}} \in [100, 250]$ mm; width $w \in [50, 250]$ mm; spoke count $n_s \in \{6, \ldots, 28\}$; wall thickness $t_w \in [2, 10]$ mm; cross angle $\theta_c \in [30°, 80°]$.

Group 2 — Material distribution (genes 6–13): each of eight $45°$-sector segments is assigned one of six FFF polymers (TPU 95A, TPU 85A, TPU 75D, Nylon PA12, ABS, PLA), $E \in [25, 3500]$ MPa. Table 1 summarises the material catalogue.

Group 3 — Pantographic configuration (genes 14–17): spoke layers $n_l \in \{1, \ldots, 15\}$; spoke thickness $t_s \in [2, 12]$ mm; hub angle $\theta_h \in [0°, 60°]$; tread/rim factor $f_{\text{tr}} \in [1.0, 4.0]$.

Table 1: Material catalogue: six FFF-printable polymers

Material	E **(MPa)**	σ_y **(MPa)**	ν	ρ **(g/cm^3)**	**Model**
TPU 95A	50	25	0.47	1.15	Neo-Hookean
TPU 85A	25	15	0.47	1.12	Neo-Hookean
TPU 75D	200	40	0.45	1.20	Neo-Hookean
Nylon PA12	1 800	45	0.40	1.01	Linear elastic
ABS	2 200	40	0.38	1.05	Linear elastic
PLA	3 500	55	0.36	1.25	Linear elastic

Four feasibility constraints are enforced: $d_{\text{rim}} < 0.9\, d_{\text{ext}}$; $n_s \geq 6$; $m < 5.0\,\text{kg}$; $\text{FS} \geq 1.5$.

3 THE GAD FRAMEWORK

3.1 Pipeline overview

The GAD pipeline has five sequential modules; the 3D FEM module is invoked on-demand by Surrogate Refresh. A genome is decoded into Phase 1 (dataset generation by LHS and 2D FEA), which trains Phase 2 (the GAT surrogate), which in turn drives Phase 3 (NSGA-III). Every 15 generations, Surrogate Refresh calls Phase 4 (3D FEM validation) on the top-5 Pareto candidates and feeds the correction factors back into Phase 3. Phase 5 exports STL geometry and a print guide. The complete pipeline runs in 4–12 hours on a Ryzen 5 7600X with an AMD RX 9070 (497 220 evaluations in the extended study).

3.2 Phase 1 — Dataset generation

N genome–property pairs are generated via Latin Hypercube Sampling, filtered by the four feasibility constraints. Five deterministic baselines (homogeneous TPU, graded, flexible–rigid, nylon-core, ABS-rim) enrich the dataset. Multi-process parallelism (8 workers) and incremental checkpointing keep generation tractable.

3.3 Phase 2 — Graph attention network surrogate

Each wheel is modelled as a directed attributed graph $G = (V, E)$. Nodes represent structural connection points (hub, rim, tread, spoke crossings) with 6D feature vectors $(r, \theta, z, \text{type}, n_l, w_s)$. Edges represent spokes with 8D vectors $(\ell, \theta_c, t_s, \text{type}, n_l, E/1000, \rho, \sigma_y)$.

The architecture comprises (i) node and edge encoders (MLP to 256 dimensions); (ii) three GAT layers ($256 \to 128$, 2 heads; $256 \to 128$, 2 heads; $256 \to 256$, 1 head) [2, 3]; (iii) global mean pooling (256 dimensions); (iv) a genome encoder $18 \to 64$ and an FEA feature encoder $22 \to 64$; (v) concatenation to 384 dimensions followed by 17 regression heads. An ensemble of five models is trained with AdamW, ReduceLROnPlateau, and early stopping (patience 25); transfer learning from a pre-trained pantograph surrogate accelerates convergence.

3.4 Phase 3 — NSGA-III with Surrogate Refresh

NSGA-III [4] optimizes ten objectives simultaneously (Table 2) with a population of 100 over 200 generations ($1\,000 \times 500$ in the extended study), using Das–Dennis reference directions [1].

Table 2: Ten optimisation objectives

#	Objective	Direction
1	Vertical stiffness k_v	Max
2	Lateral stiffness k_l	Max
3	Torsional stiffness k_t	Max
4	Safety factor FS	Max
5	Fatigue life N_f	Max
6	Shape-recovery ratio	Max
7	Stored elastic energy	Max
8	Mass m	Min
9	Maximum deflection δ	Min
10	Print complexity	Min

Surrogate Refresh. Every 15 generations the top-5 Pareto-front candidates are re-evaluated with 3D FEM, and the correction factors

$$f_k = \frac{k_{\text{FEM3D}}}{k_{\text{FEA2D}}}, \quad f_\delta = \frac{\delta_{\text{FEM3D}}}{\delta_{\text{FEA2D}}}, \quad f_\sigma = \frac{\sigma_{\text{FEM3D}}}{\sigma_{\text{FEA2D}}} \tag{2}$$

are updated via an exponential moving average with $\alpha \in [0.15, 0.60]$, clamped to $[0.1, 5.0]$ before the update (preventing the $96\times$ unclamped factor observed in an earlier version) [5]. The top-20 Pareto candidates receive a final full 3D FEM evaluation at run end.

3.5 Phase 4 — 3D FEM high-fidelity analysis

The FEM module uses scikit-fem v12.0.1 with TetP2 elements (10-node second-order tetrahedra) [8], meshed at $h = 1.5\,\text{mm}$ with fully clamped rim nodes. The design load $F_{\text{vert}} = 1{,}584\,\text{N}$ is applied as a distributed traction on the lower tread nodes; $\mathbf{Ku} = \mathbf{f}$ is solved with `scipy.sparse.linalg.spsolve`. Four separate runs (one per constituent polymer) yield the worst-case von Mises stress and safety factor per material.

4 ANALYTICAL 2D FEA MODEL

Fast fitness evaluation inside NSGA-III uses a Timoshenko-beam model ($< 10\,\text{ms}$ per evaluation).

Spoke stiffness. Axial and bending contributions combine as

$$k_{\text{axial}} = \frac{EA_s}{L_s}\sin^2\theta_c, \qquad k_{\text{bend}} = \frac{12EI_w}{L_s^3}\cos^2\theta_c, \tag{3}$$

with each additional layer contributing 60 % of the first layer's stiffness. The total stiffness follows a Voigt saturation model

$$k_{\text{total}} = \frac{K_{\text{MAX}}\, k_s\, n_s}{K_{\text{MAX}} + k_s\, n_s}, \qquad K_{\text{MAX}} = 200 + 600\,(t_s/12)\ \text{N/mm}. \tag{4}$$

K_{MAX} is an empirical cap, not a derived quantity; its consequences are quantified in Section 6.3.

Stress, lateral and torsional stiffness, fatigue, shape memory. Within this analytical model only, the notch effect is estimated with Neuber's factor $K_t = 1 + 2\sqrt{t_s/(2\rho)}$, $\rho = \max(0.5, 0.3t_w)$ [10]. Lateral and torsional stiffness follow $k_l = k_v \tan(\theta_c) \times 0.4$ and $k_t = GJ_p n_s/L_s/1000$ N·m/rad. Fatigue follows Goodman–Basquin [11] with N_f capped at 10^{12}. TPU sectors use a Neo-Hookean strain energy at $\lambda_{\max} = 1.30$ [12]; the recovery ratio is $\text{clip}(0.99 - 0.07\sigma_y/50,\ 0.92,\ 0.99)$.

5 OPTIMAL CANDIDATE: ID 79

5.1 Genome and physical parameters

The extended run (1 000 individuals, 500 generations, 497 220 FEA evaluations, 22 387 s) produced a 103-solution Pareto front. Table 3 reports the decoded genome of Candidate ID 79, the peak-stiffness solution.

Key design choices: spoke thickness is driven to its upper bound (11.98 mm), which is the only genome variable entering K_{MAX} in Eq. (4); TPU 85A occupies sector 4 (the ground-contact zone) for impact absorption; rigid polymers alternate at 90° for symmetric stiffness. Derived spoke geometry: $L_s = 67.75$ mm, $A_s = 453.1$ mm^2, $I_w = 5{,}416$ mm^4.

5.2 Multi-objective performance

Table 4 reports all ten objectives for Candidate ID 79.

[a] Obtained from the Neo-Hookean sector model at $\lambda_{\max} = 1.30$ (Section 5), not from the linear stiffness result; the two are not interchangeable.

[b] Consistent with $\delta = F_{\text{vert}}/k_v = 1584/796.6$ kN/m.

6 RESULTS

6.1 3D FEM stress analysis

The FEM mesh for ID 79 has 59 486 nodes and 309 174 TetP2 elements (1 328 466 DOF), solved in 125.0 s. Table 5 reports worst-case results under $F_{\text{vert}} = 1{,}584$ N ($n_c = 4$ active spokes).

The governing case is TPU 85A ($\text{FS} = 20.06$), above the $\text{FS} \geq 1.5$ requirement, with zero plastic strain in all materials. The two right-hand columns are ratios of the local von Mises stress

Table 3: Decoded genome of Candidate ID 79

Group	Parameter	Gene g_i	Value	Unit
Geometry	Outer diameter d_{ext}	0.0529	310.6	mm
	Rim diameter d_{rim}	0.0591	108.9	mm
	Width w	0.0026	50.5	mm
	Spoke count n_s	0.9898	28	–
	Wall thickness t_w	0.9323	9.46	mm
	Cross angle θ_c	0.4319	51.6	°
Materials	Sector 0 (0°–45°)	0.4073	TPU 75D	–
	Sector 1 (45°–90°)	0.9587	PLA	–
	Sector 2 (90°–135°)	0.7667	ABS	–
	Sector 3 (135°–180°)	0.3091	TPU 75D	–
	Sector 4 (180°–225°)	0.1071	**TPU 85A**	–
	Sector 5 (225°–270°)	0.9261	PLA	–
	Sector 6 (270°–315°)	0.8021	ABS	–
	Sector 7 (315°–360°)	0.3621	TPU 75D	–
Pantographic	Spoke layers n_l	0.0460	2	–
	Spoke thickness t_s	0.9976	11.98	mm
	Hub angle θ_h	0.7565	45.4	°
	Tread/rim factor	0.1960	1.59	–

Table 4: Ten optimisation objectives for Candidate ID 79

Objective	Value	Unit	Direction
Vertical stiffness k_v	796.6	kN/m	↑
Lateral stiffness k_l	402.0	kN/m	↑
Torsional stiffness k_t	13 077	N·m/rad	↑
Safety factor FS	20.06	–	↑
Fatigue life N_f	$> 10^{12}$	cycles	↑
Shape-recovery ratio	0.969	–	↑
Stored elastic energy[a]	7 318.7	mJ	↑
Mass m	3.346	kg	↓
Maximum deflection δ[b]	1.99	mm	↓
Print complexity	0.667	–	↓
Structural efficiency k_v/m	237.1	(kN/m)/kg	–

to the nominal stress $\sigma_{\text{nom}} = 0.685\,\text{MPa}$ measured in the same solution; a value below unity at

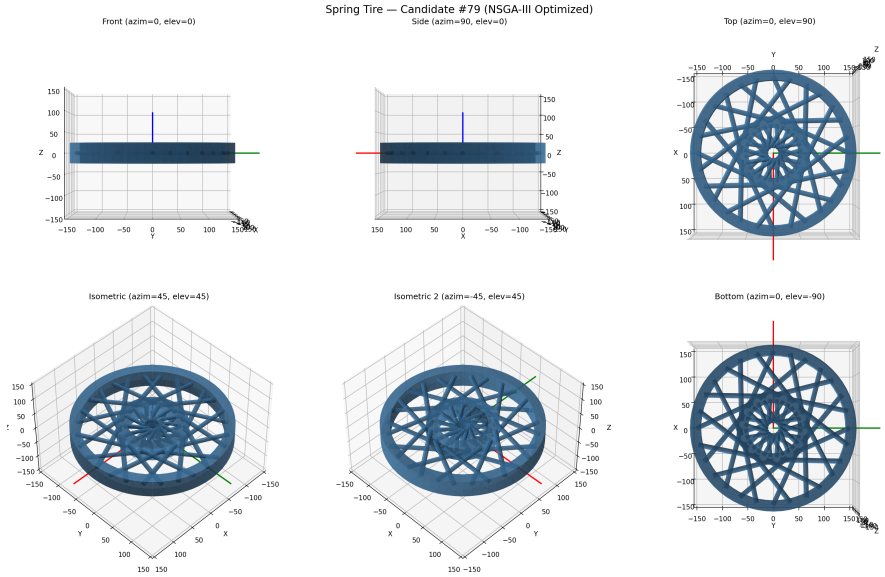


Figure 2: Six orthographic and isometric views of Candidate ID 79 (Spring Tire, NSGA-III optimised). $d_{\text{ext}} = 310.6\,\text{mm}$, $w = 50.5\,\text{mm}$, $n_s = 28$ spokes, $\theta_c = 51.6°$, dual pantographic layers.

Table 5: 3D FEM results by material, worst-case analysis (ID 79, $h = 1.5$ mm, 309 174 TetP2 elements; solution time per run in parentheses)

Material	E (MPa)	σ_y (MPa)	σ_{vm} (MPa)	$\sigma_{\text{vm}}/\sigma_{\text{nom}}$ (rim)	$\sigma_{\text{tread}}/\sigma_{\text{nom}}$ (tread)	**FS**
TPU 85A	25	15	0.748	1.092	0.784	**20.06** (125.0 s)
TPU 75D	200	40	0.789	1.151	0.824	50.72 (100.4 s)
ABS	2 200	40	0.874	1.276	0.938	45.75 (77.7 s)
PLA	3 500	55	0.893	1.303	0.965	61.61 (71.1 s)

the tread simply indicates that the tread–spoke junction is less loaded than the nominal section, and does not correspond to a notch factor. Regarding surrogate accuracy, the 2D FEA model gave $\sigma_{\text{vm}} = 1.134\,\text{MPa}$ — the 3D value is 34 % lower, i.e. $f_\sigma = 0.66$ — and $k_v = 793.2\,\text{kN/m}$, 0.4 % below the 3D value, i.e. $f_k \approx 1.004$.

6.2 Mesh convergence and numerical uncertainty

Table 6 reports the three-mesh study.

Richardson extrapolation over these three meshes returns an apparent order $p = 1.11$ and $\text{GCI} = 305\,\%$. Two observations show that this triplet must not be read as an uncertainty band. First, the extrapolated value is $\sigma_{\text{ext}} \approx -1.1\,\text{MPa}$, which is not physically admissible for a von Mises stress; the sequence therefore lies outside the asymptotic range in which Richardson ex-

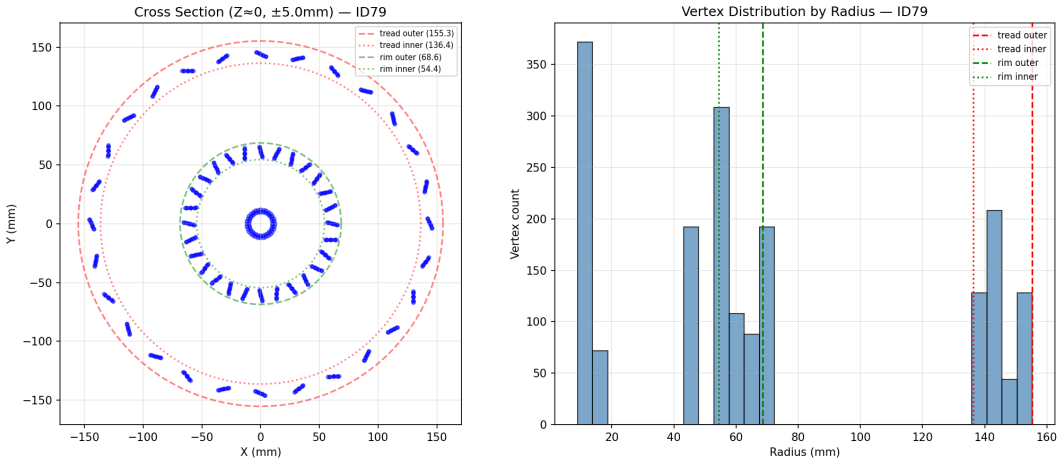


Figure 3: Cross-section geometry of ID 79 ($Z \approx 0$, ± 5 mm slice). Left: tread outer (155.3 mm), tread inner (136.4 mm), rim outer (68.6 mm), rim inner (54.4 mm). Right: vertex distribution by radius, confirming spoke placement within the spoke region ($r \in [55, 70]$ mm).

Table 6: Mesh study for ID 79, TPU 85A

h (mm)	Nodes	Elements	σ_{vm} (MPa)	FS
4.0	4 759	20 559	4.600	3.26
2.5	15 151	72 493	2.142	7.00
1.5	59 486	309 174	0.748	20.06

trapolation is defined [13]. Second, the stress falls by a factor of six under refinement, whereas a geometric stress singularity — the mechanism proposed in earlier versions of this analysis — produces the opposite trend, an unbounded increase. The more likely explanation is that the two coarser meshes do not resolve the spoke cross-section: at $h = 4.0$ and 2.5 mm there are approximately three and five elements across the 11.98 mm spoke thickness, so their stresses are discretisation artefacts rather than converged values.

A meaningful convergence study therefore requires refining from $h = 1.5$ mm downward, with fillets ($r = 0.3t_w$) introduced at the rim–spoke junction. Until that study is available, $\sigma_{\text{vm}} = 0.748$ MPa at $h = 1.5$ mm must be treated as unverified and the associated safety factor as indicative only. The stiffness results are independent of this mesh, as they come from the Timoshenko model of Section 5 (uncertainty $\pm$30–40 %).

6.3 Stiffness, saturation, buckling and shape memory

The stiffness decomposition gives $k_{\text{axial}} = 6{,}173.8$ N/mm (98 %) and $k_{\text{bend}} = 121.2$ N/mm (2 %), for a design deflection $\delta = 1.99$ mm.

Saturation of the stiffness model. With $k_s = k_{\text{axial}} + k_{\text{bend}} = 6{,}295$ N/mm and $n_s = 28$,

the product $k_s n_s = 176{,}259\,\text{N/mm}$ exceeds $K_{\text{MAX}} = 798.8\,\text{N/mm}$ by more than two orders of magnitude, so Eq. (4) returns $k_v = 0.995\,K_{\text{MAX}}$. The reported vertical stiffness is thus governed almost entirely by the empirical cap $K_{\text{MAX}} = 200 + 600(t_s/12)$, whose only argument is the spoke thickness. Reducing n_s from 28 to 6 lowers k_v by just 1.6 %. Two consequences follow and are carried into Section 7: objective 1 does not discriminate between spoke counts, so the value $n_s = 28$ selected by the optimiser is not evidence that many spokes are mechanically preferable; and calibrating or replacing K_{MAX} against physical compression tests is the single most important validation step for this model.

Johnson–Euler buckling places the spokes in the short-column regime ($\lambda = 19.6 < \lambda_{\text{tr}} = 27.03$), with $P_{cr} = 13{,}568\,\text{N}$ and a buckling safety factor of 239.8, indicating no risk. Neo-Hookean shape recovery at $\lambda_{\max} = 1.30$ gives a composite system-level ratio of 96.9 % with 7 318.7 mJ of stored elastic energy.

6.4 Fatigue analysis

Applying the Goodman criterion with $R = 0.1$ to TPU 85A,

$$\frac{\sigma_a}{S_e} + \frac{\sigma_m}{S_{ut}} = \frac{0.337}{6.825} + \frac{0.411}{19.5} = 0.070 \ll 1. \tag{5}$$

All four materials achieve Goodman ratios below 0.031 (Table 7), consistent with $N_f > 10^{12}$ cycles under the Basquin extrapolation.

Table 7: Fatigue analysis for Candidate ID 79, with σ_{vm} taken from the 3D FEM solution

Material	S_e (MPa)	S_{ut} (MPa)	σ_a (MPa)	σ_m (MPa)	**Goodman** ratio	FS_{fat}
TPU 85A	6.825	19.5	0.337	0.411	0.070	14.2
TPU 75D	18.20	52.0	0.355	0.433	0.028	35.9
ABS	18.20	52.0	0.393	0.480	0.031	32.4
PLA	25.03	71.5	0.402	0.491	0.023	43.6

FS_{fat} is the reciprocal of the Goodman ratio. It is a distinct quantity from the static safety factor of Table 5 and is numerically smaller; both remain well above unity.

6.5 Thermal durability and UV degradation

Thermal stresses ($\sigma_{\text{th}} = E\alpha\Delta T f_{\text{rest}}$, $f_{\text{rest}} = 0.15$) give $\text{FS}_{\text{th}} = 1.16$ under a nominal $\Delta T = 80\,^{\circ}\text{C}$; the safety factor drops below unity at polar winter ($\Delta T = 145\,^{\circ}\text{C}$) once $f_{\text{rest}} \geq 0.27$, which would require a full 3D thermal analysis. The thermal fatigue margin is $N_{f,\text{th}} \approx 10^9$ cycles against 5 840 mission cycles.

Martian UV (2–3× Earth, with a higher UV-C component) degrades stiffness exponentially [14]. Counter-intuitively, the structural safety factor increases over five years, because the

stiffness reduction lowers the transmitted stress faster than the yield limit degrades (Table 8). This result must be read with caution: UV exposure also embrittles the polymer (fracture-toughness degradation), a mode the present model does not capture, so the trend in Table 8 should not be taken as evidence of improving durability.

Table 8: Structural safety factor under UV degradation for Candidate ID 79

Material	0.5 yr	1 yr	2 yr	3 yr	5 yr
TPU 85A	20.3	20.5	20.9	21.4	22.3
TPU 75D	54.1	54.8	56.1	57.5	60.3
ABS	55.4	57.6	62.1	66.9	77.8
PLA	76.8	80.3	87.9	96.3	115.3

6.6 Surrogate accuracy and computational cost

After Surrogate Refresh, the GAT ensemble predictions agree to within $\pm 1\,\%$ (stiffness) and $\pm 5\,\%$ (von Mises) on Pareto-front candidates, at a 100-fold speed-up ($\sim$5 ms versus $\sim$120 s per evaluation).

7 LIMITATIONS AND FUTURE WORK

All results derive from open-source simulations (scikit-fem, NumPy/SciPy, PyTorch Geometric); no commercial FEM cross-validation was performed. The explicit limitations are as follows.

Mesh. The three-mesh sequence is not in the asymptotic range (Section 6.2): Richardson extrapolation returns a non-physical negative stress, so no verified numerical uncertainty can be quoted for σ_{vm}. A study refining below $h = 1.5\,\mathrm{mm}$, with fillets at the rim–spoke junction, is required before the reported safety factors can be treated as verified.

Stiffness. The Timoshenko–Voigt model carries $\pm$30–40 % uncertainty, and k_v is 99.5 % determined by the empirical cap K_{MAX} rather than by the spoke mechanics (Section 6.3). Objective 1 consequently does not discriminate between spoke counts, and the optimiser's preference for $n_s = 28$ should not be interpreted mechanically. Prototype compression testing is the priority validation, both to calibrate K_{MAX} and to establish whether the saturation form is appropriate at all.

FEM. Single-material runs only; true multi-material FEM with interface contact is needed. Figure 4 shows that the current boolean-union geometry produces no explicit interface edges between rim, spokes and tread, which is a prerequisite for such an analysis.

FDM anisotropy. A 15–30 % reduction in E along the build direction is not captured by the isotropic model.

Fatigue data. Basquin constants are taken from bulk literature; FDM parts typically show 20–50 % lower life, so the $N_f > 10^{12}$ figure is an extrapolation rather than a measurement.

UV and thermal. Rates are extrapolated from LEO data; fracture toughness is not modelled, and the f_{rest} range 0.05–0.40 requires a 3D thermal analysis.

No prototype. All results are design-exploration outputs, not engineering specifications.

Future work: mesh refinement below $h = 1.5\,\text{mm}$ with fillets; multi-material FEM with interface contact; experimental calibration of K_{MAX}; dynamic modal analysis; topology optimisation within sectors; and extension of the framework to a segmented outer-ring wheel architecture evaluated by multibody rolling simulation, where the objectives are energetic rather than static.

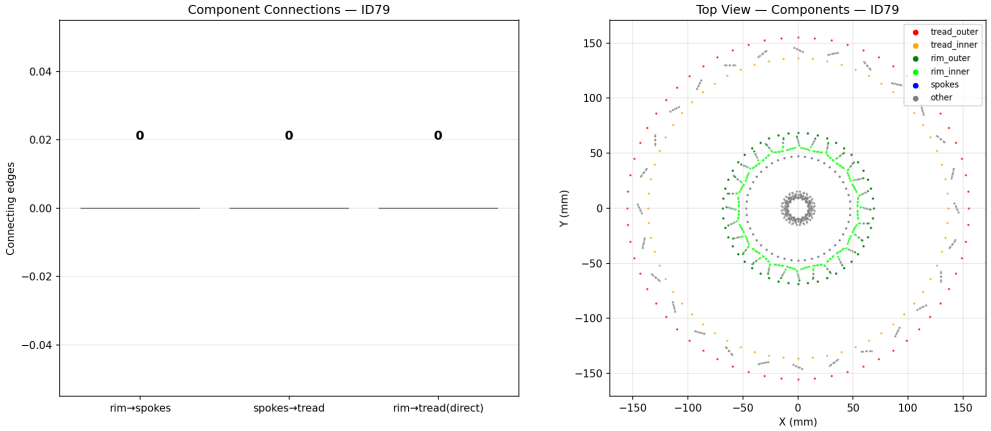


Figure 4: Component connectivity analysis of ID 79. Left: zero connecting edges detected between rim–spokes, spokes–tread and rim–tread, indicating that the current boolean-union geometry does not generate explicit interface edges in the graph representation. Right: top-view node classification by component. This motivates future work on multi-material interface FEM.

8 CONCLUSIONS

This paper introduced Graph-Assisted Design (GAD), an automated pipeline for multi-objective structural optimisation of compliant multi-material airless wheels. The main contributions are:

1. An 18-gene genome encoding geometry, pantographic spokes, and a heterogeneous eight-sector material distribution.

2. A GAT ensemble with three encoders reaching $\pm 1\,\%$ stiffness error after Surrogate Refresh, at a 100-fold speed-up over 3D FEM.

3. The Surrogate Refresh mechanism itself, with EMA correction and clamping to $[0.1, 5.0]$, which bounds prediction drift and prevents the optimiser from exploiting surrogate error.

4. Candidate ID 79: $k_v = 796.6\,\text{kN/m}$, $\text{FS} = 20.06$, Goodman ratio 0.070, recovery 96.9 %, mass 3.346 kg, evaluated on a 1 328 466-DOF TetP2 solution.

5. An explicit uncertainty audit which establishes that two of these figures are not yet verified: the mesh sequence is outside the asymptotic range, and the vertical stiffness is governed by an empirical cap rather than by the spoke mechanics. Both are stated as open items rather than resolved ones.

The GAD framework is domain-agnostic, and a preliminary case study on compliant robotic-arm topology confirms its generalisability.

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